

SFT data distillation framework for OR modeling

Generated Questions-Abstract Seed

Seed Samples

High-quality
Modeling QA



- Original Problems
- Answers
- Reference Code
- Data Files / LP Files

L1 Canonical IR

Target Contract,
Inline Text, Risks



- Target Contract
- Static Code Features
- Labels
- Objective Hint

L2 Semantic IR

Objective, Variables,
Constraints



- Objective
- Sets / Parameters
- Variables
- Constraints

L3 Synthetic IR

New Modeling Task



- New Domain
- New Entities & Data
- New Objective
- New Constraints

Iterable Seed Pool



Problem Rendering

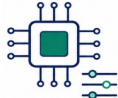
DP / DT / DPS Problem



- Inline Data
- Tables
- External Files

Answer Rendering

Natural Language & Gurobi



- <think> <model>
- <python>
- Gurobi model
- Print Objective Value

Quality Inspection

Executable and Accepted



- Code Executes
- Contract Matched
- Output Verified
- Pack to SFT Row

Generated Full Problem

Generated problem:

A logistics company needs to transport packages from a depot to a city using vans and trucks. Each van trip carries **12 packages** and takes **18 minutes**; each truck trip carries **30** and takes **40 minutes**. At least **5 truck trips** are required, van trips must be **at least twice** truck trips, and at least **400 packages** must be delivered. Minimize total time. Validated objective: 568.0

Abstraction

Abstract Modeling Seed

Objective: $\min 18V + 40T$.
Variables: $V \text{ integer} \geq 0, T \text{ integer} \geq 5$.
Constraints: $12V + 30T \geq 400; V \geq 2T$.
Surface wording is discarded;
modeling structure is kept as reusable seed.